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Phase separation and crystallization effects on the structure and durability of molybdenum borosilicate glass

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ABSTRACT

The effects of phase separation by nucleation and growth as well as those of crystallization on the structure and the chemical durability of sodium borosilicate glasses enriched in molybdenum oxide were investigated. Increasing concentrations of MoO_3 (0 to 10% molar) were added to glasses which were homogeneous ($MoO_3 \leq 1.4\%$ molar) or with crystallizations and separated phases ($MoO_3 > 1.4\%$ molar). Their structure was determined by ^{11}B, ^{23}Na, ^{29}Si and ^{95}Mo MAS NMR, and ^{23}Na MQMAS NMR, coupled with DFT calculations of NMR parameters. Their microstructure and morphology were characterized by X-ray diffraction and by electronic microscopy. These complementary approaches pointed out a decrease in the number of non-bridging oxygens (NBOs) within the vitreous matrix, first initiated by an increase in polymerization for glasses with a low molybdenum oxide content (<2 % molar), and combined with the decrease in the number of tetrahedral boron for higher contents. Correlations between the chemical durability, the structure and the microstructure of the glasses were then obtained at low and high dissolution reaction progress. The modifications to the chemical composition and to the structure of the surrounding glass induced by increasing MoO_3 strongly influenced the matrix durability. Moreover, the nature and the morphology of the separated phases influence the dissolution kinetics by modifying the solution chemistry.

1. Introduction

The phase separation and crystallization processes within glasses modify their physico-chemical properties (viscosity, thermal dilatation, crystallization behavior, chemical durability) [1]. Controlling phase separation enables a wide variety of heterogeneous materials to be fabricated, opening up a broad range of industrial applications (Pyrex, Vycor, opal glass, porous glass) [2–4]. While such mechanisms may be considered to be advantageous in vitroceramics, they can however deteriorate the quality of the material if they are not controlled. For example, phase separation obtained by a heat treatment of nucleation and growth [1,5–7] or by the addition of nucleating agents [8,9] to sodium borosilicate glasses modifies glass behavior in an aqueous medium.

Depending on the nature, the composition and the morphology of the separated phases formed, demixed glasses can be more or less durable compared to the initial homogeneous glasses. Ternary glass compositions (SiO_2–Na_2O–B_2O_3) with a miscibility gap, for example, enable demixed glass compositions (e.g., Vycor), which are easily leached. This type of phase separation is on the other hand to be avoided when highly durable glass is required, for radioactive waste conditioning glass for geological disposal [10,11].

Borosilicate glasses demixed by the addition of molybdenum oxide represent a model simplified system for investigating heterogeneous glasses. Molybdenum is a fission product coming from spent fuel reprocessing solutions, and is weakly soluble in glass [12–15]. When the content is higher than its solubility limit, processes of phase separation and of crystallization lead to the formation of heterogeneous glasses [16]. Several authors have pointed out the influence of molybdenum oxide on the structure and microstructure of sodium borosilicate glasses. Initiated by a liquid–liquid phase separation at high temperature, separated phases enriched in molybdenum oxide crystallize during cooling [14,17,18], giving partially crystallized heterogeneous material containing alkali and alkaline-earth molybdates.

It has been suggested that these mechanisms occur at an atomic scale by the clustering of several isolated molybdate units. Molybdenum, mainly under its oxidation state + VI in glasses prepared in oxidizing conditions [16,19–21], is found in the form of MoO_4^{2-} tetrahedral units distributed in depolymerized areas of glass with charge-compensating cations (for example alkalis and alkaline-earths). This

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molybdenum structural environment leads to a repolymerization of the remaining silicate network [22,23] following the equation:

$$MoO_3 + \left(2Q^3, Ca^{2+}\,\text{or}\,2Na^+\right) \rightarrow \left(MoO_4^{2-}, Ca^{2+}\,\text{or}\,2Na^+\right) + 2Q^4.$$

In soda-lime borosilicate glass, the nature of the separated phases and of the crystalline phases depends on the charge compensator cation of the molybdates. It has been shown that the quantity of Na^+ remaining in the glass network after compensation for the BO_4^- units governs the local environment of the molybdate units and thus the nature of the phases formed during the process of phase separation and crystallization. The decrease in the number of "free" Na^+ ions – induced by the gradual increase of CaO and B_2O_3 content – leads to the formation of partially crystallized glasses containing a mixture of Na_2MoO_4 and $CaMoO_4$ phases, or $CaMoO_4$ only, depending on the glass composition [18,22,24]. Greater understanding of the chemical durability properties of this type of glass can be addressed by investigations of its structural and morphological changes.

In this work, a series of borosilicate glasses with varying molybdenum oxide contents were studied. The glasses contained enough sodium to compensate $[BO_4]^-$ and $[MoO_4]^{2-}$ units, and to form sodium molybdates during the phase separation and crystallization processes. The addition of increasing concentrations of MoO_3 (0 to 10% molar) enabled the fabrication of either glasses which were homogeneous at nanometric scale (MoO_3 content <1.4% molar), or glasses containing separated phases at a microscopic or macroscopic scale when the molybdenum oxide content was high (>2% molar). Using complementary approaches with solid-state high resolution nuclear magnetic resonance (^{11}B, ^{23}Na, ^{29}Si, ^{95}Mo), high resolution transmission electron microscopy (HRTEM) and X-ray diffraction with Rietveld refinement, it becomes possible to give a detailed description of the structural, morphological and microstructural evolution of glasses. All the glasses were leached in deionized water at pH 9 and at 90 °C, following different alteration regimes (short and medium term). These approaches give further insights into the aqueous reactivity of homogeneous vitreous matrices containing varying amounts of phases whose chemical and/or geometric nature differ.

2. Experimental procedure

2.1. Glass preparation

A series of molybdenum-bearing sodium borosilicate glasses (*x*Mo) in which the molybdenum oxide proportionally replaced other oxides following the formula $((100x)\ 63SiO_2 - 17B_2O_3 - 20Na_2O) - xMoO_3$ with $x = 0, 0.5, 0.8, 2, 3, 4, 5$ and 10 were investigated. The glasses were prepared by mixing appropriate quantities of the precursors SiO_2, H_3BO_3, Na_2CO_3 and MoO_3. The powder mixtures were then melted at 1250 °C for 3 h in Pt–Rh crucibles, then poured and cooled between two copper plates with a cooling rate between $10^2\ °C\cdot min^{-1}$ and $10^3\ °C\cdot min^{-1}$ [16,25]. The glass chemical compositions were checked by alkaline fusion followed by an analysis with inductively coupled plasma-optical emission spectroscopy (ICP-OES). The analyses gave the theoretical composition at less than 0.5% of nominal compositions listed in Table 1.

Table 1
Nominal composition of the *x*Mo glass series in % molar.

Glass	0Mo	0.5Mo	0.8Mo	2Mo	3Mo	4Mo	5Mo	10Mo
SiO_2	63.5	63.2	63.0	62.2	61.1	61.0	60.3	57.1
B_2O_3	16.9	16.8	16.8	16.6	16.5	16.2	16.1	15.2
Na_2O	19.6	19.5	19.4	19.2	19.4	18.8	18.6	17.6
MoO_3	–	0.5	0.8	2.0	3.0	4.0	5.0	10.0

2.2. Solid characterization methods

2.2.1. Powder X-ray diffraction (PXRD)

After cooling, all the glass samples were characterized by powder X-ray diffraction (PXRD) using the PANalytical X'Pert Pro diffractometer equipped with a solid detector X'Celerator and using Cu $K\alpha_{1,2}$ radiation ($\lambda = 1.54184$ Å). All the PXRD patterns were collected in the Bragg–Brentano geometry in the angular range of $10° < 2\theta < 80°$, with a step size $\Delta 2\theta = 0.017°$ and total counting time of at least 5 h. PXRD of pure silicon (Si) was collected in the same condition and used to extract the instrumental function. Moreover, a known amount of Si was also mixed to each phase and used as internal standard to quantify the amount of crystallized phase vs amorphous. Besides the Si, the crystalline phases present in each PXRD were identified by comparison with the International Center for Diffraction Data.

After that, all the PXRD patterns were refined by the Rietveld method using the "Fullprof_suite" program [26]. During the refinement, different parameters were allowed to vary: zero shift, scale factors, preferred orientation, asymmetry parameters, lattice parameters, a global thermal displacement for each phase, and intrinsic microstructure parameters to extract size effect.

The solubility limit of the molybdenum oxide and the presence of a low crystalline phase content in the glasses were also checked by dissolution heat measurement with a CALSOL calorimeter [27].

2.2.2. Solid-state nuclear magnetic resonance

The spectra were collected on a Bruker Avance II 500WB spectrometer operating at a magnetic field of 11.72 T. ^{11}B, ^{23}Na and ^{29}Si MAS NMR spectra were acquired using a Bruker CPMAS probe (with ZrO_2 rotor, outer diameter (OD) 4 mm) at a sample rotation frequency of 14 kHz for all nuclei. For ^{95}Mo, a Bruker CPMAS probe with 7 mm (OD) ZrO_2 rotor was used at a spinning frequency of 6 kHz.

2.2.2.1. ^{11}B MAS NMR. To ensure spectrum quantitativeness, a short pulse of 1 μs (tip angle of π/12) [30] and a recycle delay of 2 s were used. Chemical shifts were referenced to an external sample of 1 M boric acid solution (19.6 ppm relative to the boron trifluoride etherate).

2.2.2.2 . ^{29}Si MAS NMR. Silicon spectra were acquired using the CPMG sequence [28], by typically accumulating 32 echoes with an echo delay of 2.4 ms between consecutive 180° pulses. The echoes were then summed and Fourier transformed to obtain the spectra shown. A 20 s recycling time was used; checks were carried out on two glass samples that no spectral deformation could be observed for longer recycling delays (200 s and 1200 s). The spectra were referenced to an external tetrakis(trimethylsilyl)silane (TKS) sample for which the highest-intensity peak is situated −9.9 ppm from that of TMS.

2.2.2.3. ^{23}Na MAS and MQMAS NMR. A short pulse (π/12 with a pulse width of 1 μs) and a recycle delay of 1 s were used for the MAS spectra. ^{23}Na MQMAS spectra were acquired with a RIACT-II two-pulse sequence [29] with optimized excitation and reconversion pulses of $p_1 = 7$ μs and $p_2 = 18$ μs respectively (*rf* field strength of 80 kHz, the second pulse length matched one quarter of the rotation period), and 64 t_1 increments of 10 μs were acquired. ^{23}Na chemical shifts were referenced to an external sample of 1 M NaCl (0 ppm).

2.2.2.4. ^{95}Mo MAS NMR. A spin echo pulse sequence with a rotor-synchronized echo delay of one rotation period was used with a recycle delay of 1 s. Because of its low sensitivity (natural abundance of 15.9%, Larmor frequency of 32.5 MHz), typically 100,000–200,000 scans were accumulated. The moderate value of its nuclear quadrupolar moment ($Q = -2.2 \cdot 10^{-30}\ m^2$), combined with its nuclear spin ($I = 5/2$), enabled the use of bigger 7 mm rotor to increase the sample volume for a better signal-to-noise ratio. ^{95}Mo chemical shifts were referenced

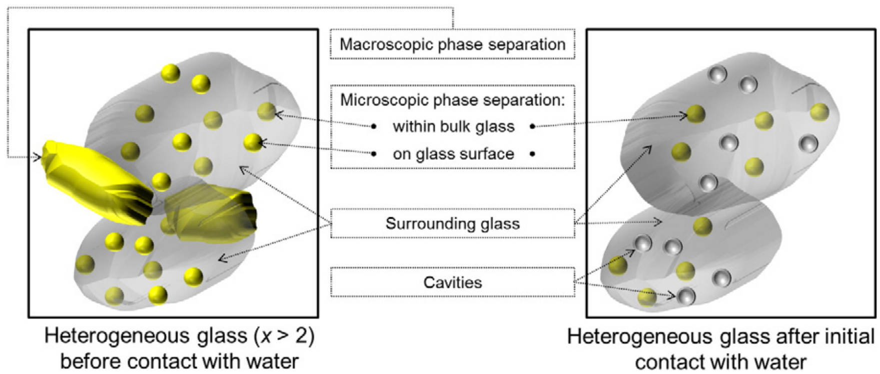

Fig. 1. Crystallized glass ($x > 2$) before and after rinsing.

to an external sample of 1 M Na_2MoO_4. Data were processed and fitted using in-house software [30].

2.2.3. Electron microscopy

Observations by scanning electron microscopy (SEM ZEISS SUPRA 55, 15 kV, EDS analysis (energy dispersive X-ray spectroscopy)) as well as by high resolution transmission electron microscopy (HRTEM FEI TITAN 80-300 with Cs corrector, spot size of approximately 9 nm) were carried out on the samples. Cross section samples for HRTEM characterization were prepared using a dual-beam focused ion beam (FIB HELIOS 600 NanoLab).

2.3. Glass alteration protocols

2.3.1. Determination of glass powder specific surface areas

Leaching was carried out on glass powder with a granulometric fraction obtained by 63–125 μm sieving. The glass powders were washed in acetone, then in absolute ethanol under ultrasound to remove the fine particles. The specific surface areas were measured by krypton adsorption (BET method [31] on a Micromeritics ASAP 2020 apparatus).

2.3.2. Determination of the initial glass alteration rate

The initial alteration rates, r_0, were measured by static experiments in perfluoroalkoxy (PFA) reactors at 90 °C. 100 mg of glass powder was introduced in 1 L of solution buffered with KOH at $pH_{90\,°C} = 9$, to minimize interdiffusion and obtain a congruent dissolution of the elements making up the glasses by favoring hydrolysis reactions. The mass of glass was therefore set to obtain *SSA*/*V* ratios (glass surface area to solution volume ratio) which were low enough ($SSA/V = 5\ m^{-1}$) to avoid any retroaction on the dissolution rate of the elements coming from the glass dissolution. The pH was checked at the end of the test to be sure it had remained stable throughout the experiment. The medium was kept under vigorous stirring, and samples were taken at regular intervals, then filtered at 0.45 μm. The leachates were analyzed by ICP-OES. Taking into account the error in the specific surface area determination and in the concentrations in solution, uncertainty for the r_0 was approximately 10% [32].

The normalized mass losses, NL_i, ($g{\cdot}m^{-2}$) were calculated with Eq. (1):

$$NL_i = \frac{C_i(t)}{SSA/V \cdot x_i} \tag{1}$$

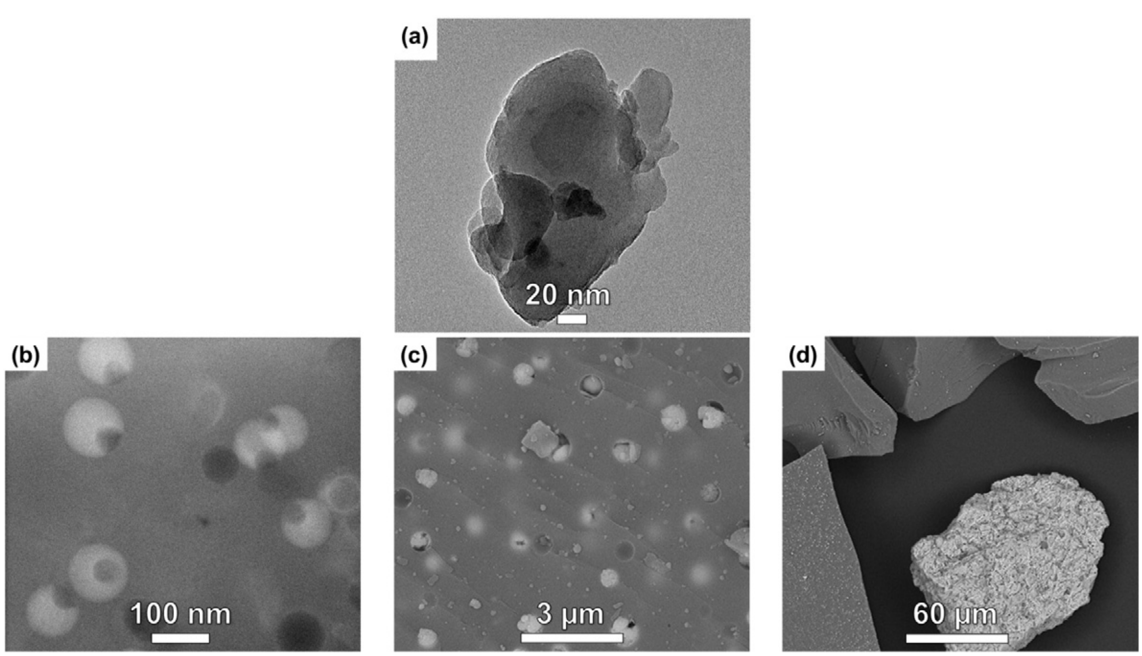

Fig. 2. (a) HRTEM image of a 1Mo ground glass grain. (b, c, d) 4Mo glass. (b) HRTEM image of a thin slice of 4Mo glass, obtained by FIB. Separated phases enriched in MoO_3 and Na_2O (lighter shade) are dispersed in a vitreous matrix (gray contrast). (c) SEM image of a glass grain containing microscopic-scale separated phases dispersed within a vitreous phase. (d) Larger scale view of a sodium molybdate phase macroscopically separated of the glass (lighter shade).

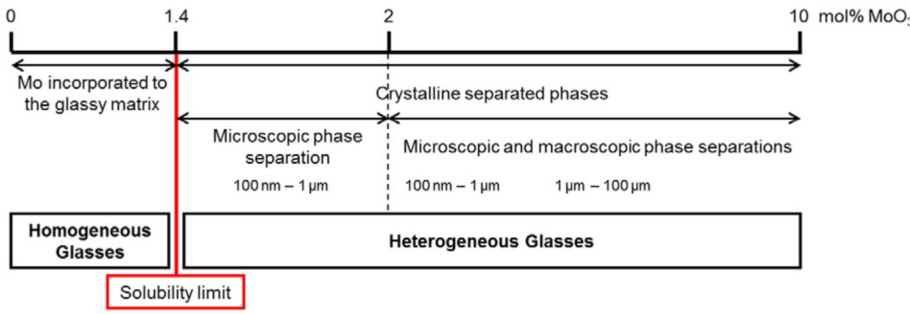

Fig. 3. Formation evolution of separated phases and of crystals depending on the molybdenum oxide content of the glass.

where $C_i(t)$ is the concentration of element i in solution at time t, x_i its mass fraction in the glass, SSA the glass specific surface area and V the solution volume.

The initial alteration rate corresponds to the slope of the curve representing $NL_i = f(t)$. To ensure measurement repeatability, each test was run at least three times.

The glasses which initially contained macroscopic separated phases were washed in deionized water at room temperature under vigorous shaking for 1 min. This washing enabled precise determination of the parameters necessary to calculate the normalized mass losses (Eq. (1)), avoiding the impact of the separated phases on the mass, the specific surface area and the glass composition. Hereafter it will be shown that the separated phases containing molybdenum, which are released in the first moments, modify the pH. However, in the initial alteration regime for which a buffer solution was used, this effect on the pH did not need to be taken into account. Once a sample of the rinsing solution had been taken, the powder was rinsed with absolute ethanol, then dried and studied by SEM to determine its morphology. With no separated phases larger than a micrometer in size, the glasses contained cavities (Fig. 1) scattered over the surface of the surrounding glass. Analysis of the washing solution concentrations was carried out to check that only the sodium molybdate had been dissolved during the rinsing protocol, without other elements being released. The specific surface area of the surrounding glass powder was measured by BET method on the washed samples. After determination of the surrounding glass composition, the washed glasses were leached under initial alteration rate conditions.

2.3.3. Rate drop regime

The rate drop regime was measured by static experiments at 90 °C in ultrapure water (18.2 MΩ·cm) on unwashed glasses, in order to not disturb the pH modifications caused by separated phase dissolution. The solution volume was 150 mL and the quantity of powder was adjusted so that the SSA/V ratio was 1500 m^{-1}. The solution samples were ultrafiltered with a cut-off threshold at 10 kDa. To take into account variations in the $m \cdot SSA/V$ ratio due to samplings of solution or of solid and to the corrosion of glass grains, normalized mass losses were calculated with Eq. (2) [33] for boron. As this element is not retained in the secondary phases or alteration layer which can form in the rate drop regime, it is considered to be the alteration tracer [34].

$$NL(\mathrm{B})_t = NL(\mathrm{B})_{t-1} + \frac{\left(C(\mathrm{B})_t - C(\mathrm{B})_{t-1}\right) \cdot V_t}{SSA \cdot m \cdot x_B} \tag{2}$$

where $C(\mathrm{B})_t$ is the concentration of boron in solution at time t, x_B its mass fraction in the glass, V_t the volume of solution sampled at time t and m the mass of glass powder.

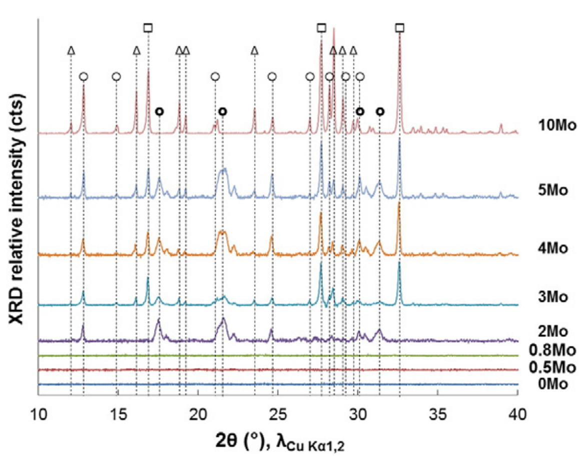

Fig. 4. PXRD patterns of the xMo series. □ Na_2MoO_4 (cubic allotropic shape, JCPDS data sheet 12-0773), ○ Na_2MoO_4, $2H_2O$ (orthorhombic, 34-0076), ⊙ Na_2MoO_4 (orthorhombic, 26-0968), and △ $Na_2Mo_2O_7$ (orthorhombic, 76-0185).

3. Results

3.1. Electron microscope studies of glass morphology

The solubility limit of molybdenum oxide in the xMo glass series was determined by a low temperature calorimetry technique during previous work [27]. This limit, determined for glass containing 1.4% molar of MoO_3, is associated with a large dissolution enthalpy variation linked to the presence of separated phases containing a Na_2MoO_4-type crystalline phase.

As shown by HRTEM (e.g., 1Mo glass, Fig. 2(a)), the glasses containing $MoO_3 < 1.4\%$ molar (0Mo, 0.5Mo, 0.8Mo, 1Mo) were homogeneous at a nanometric scale. Beyond this limit, the glasses were heterogeneous: they contained sphere-shaped phases enriched in oxygen, molybdenum and sodium, dispersed in the vitreous matrix or separated macroscopically from the glass. In the same sample, three types of morphologies and of sizes described below could be seen for the separated phases.

At a very small scale, separated phases of about 100 nm were characterized by HRTEM in the 4Mo sample (Fig. 2(b)). These contrasting light phases enriched in oxygen, molybdenum and sodium contained darker phase characteristic of cavities caused by Na_2MoO_4 crystallization during glass cooling. SEM grain analysis of this sample revealed the presence of separated phases with diameters greater than 1 µm (Fig. 2(c)), as well as other 100 µm phases which were macroscopically separated from the glass (Fig. 2(d)).

The type of morphology seen for the 4Mo glass is characteristic of all the xMo series glasses with a $MoO_3 > 2\%$ molar concentration. When the glasses had a content of $1.4\% < MoO_3 \leq 2\%$, the separated phases were dispersed within the matrix rather than being macroscopically separated from the glass. Fig. 3 summarized the formation evolution of separated phases and of crystals depending on the glass molybdenum oxide content.

Table 2
Mass composition of the crystalline phases in the *x*Mo glass series (standard deviation values are shown in brackets) and resulting average Na/Mo molar ratio.

	2Mo	3Mo	4Mo	5Mo	10Mo
Na_2MoO_4 cubic	36.2 (2.4)	28.4 (0.9)	18.4 (0.9)	22.9 (0.8)	28.6 (0.4)
Na_2MoO_4, $2H_2O$ orthorhombic	0.8 (1.1)	43.0 (0.6)	45.7 (0.7)	38.5 (0.6)	42.5 (0.4)
Na_2MoO_4 orthorhombic	63.0 (1.9)	15.8 (0.6)	20.3 (0.6)	17.4 (0.6)	–
Na_2MoO_4 total	100.0	87.2	84.4	78.8	71.1
$Na_2Mo_2O_7$	–	12.8 (0.8)	15.6 (0.8)	21.2 (0.6)	28.9 (0.4)
Na/Mo molar ratio	2.00	1.85	1.82	1.76	1.68

3.2. Powder X-ray diffraction studies of glass microstructure

The powder X-ray diffraction (PXRD) patterns for the *x*Mo series glasses viewed in Fig. 4 show the evolution of their composition depending on the MoO_3 content. In agreement with the microscopic views, when the MoO_3 content was less than 0.8% molar, the glasses don't contain any crystalline phase. The first diffraction peaks were detected as from $x = 2$. They have been attributed to the presence of Na_2MoO_4-type phases with allotropic shapes: cubic (JCPDS data sheet 12-0773), hydrated orthorhombic (34-0076) or non-hydrated (26-0968), as well as the presence of orthorhombic $Na_2Mo_2O_7$ (76-0185). The Rietveld refinement applied to these PXRD patterns enabled the determination of the composition of each of the identified phases (Table 2) depending on the quantity of molybdenum oxide added. It could thus be shown that above the solubility limit, Na_2MoO_4 (in all the allotropic forms) was the main phase present in the glasses. Its quantity gradually decreased as the $Na_2Mo_2O_7$ increased with greater amounts of MoO_3 (Fig. 5). Standard errors were given by the software “Fullprof_suite” with use of the Rietveld method which uses the least square approach to refine the calculated intensities until it matches with the experimental ones.

It should be noted that the molybdenum and sodium concentrations found after rinsing (cf 2.2.3) were identical to the Na/Mo molar ratios determined by Rietveld refinement.

3.3. Chemical composition determination for the surrounding glasses

The solubility limit determination (1.4% molar) in molybdenum oxide [27] as well as the Rietveld refinement calculation results for the Na/Mo molar ratio (Table 2) gave the total mass percentage of crystalline phases in our system and the composition of the surrounding glasses (Table 3). The total volumic percentage of crystalline phases is deduced from the mass percentage and densities of glass and molybdenum phases (Table 3).

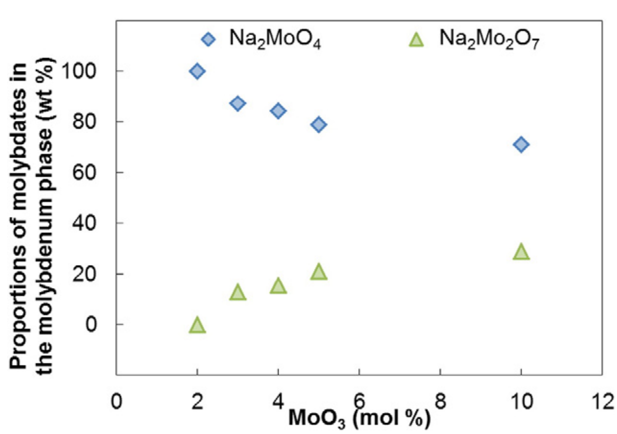

Fig. 5. Evolution of the molybdenum phase composition depending on the added MoO_3 content, ◇ wt.% Na_2MoO_4 (sum of all the allotropic forms), and Δ wt.% $Na_2Mo_2O_7$. The uncertainties are included within the width of the symbols.

However, as a macroscopic phase separation occurs in the glass with a molybdenum content higher than 2 mol%, the amount of crystalline phases contained in the glass cannot exceed 1 vol.% (amount determined for the 2Mo glass, see Table 3).

The percentage of non-bridging oxygens (%NBOs) present in the vitreous part corresponds to the quantity of sodium ions remaining after compensation from tetrahedral boron (BO_4^-) and molybdate entities (Eq. (3)). The compositions of glasses with $x \leq 0.8$ were identical to those given in Table 1.

$$\%\mathrm{NBOs} = \frac{2(\mathrm{mol\%\,Na_2O} - \mathrm{mol\%\,MoO_3} - \mathrm{mol\%\,B_2O_3} \cdot \%\mathrm{BO_4})}{\sum \mathrm{O}} \cdot 100 \quad (3)$$

3.4. Glass structure

By modifying the distribution of alkalis within the vitreous network, the addition of molybdenum oxide can cause both boron coordination changes and modifications to the silicate network polymerization. These two effects can be investigated by ^{11}B and ^{29}Si MAS NMR. Alkali molybdates can be seen by directly probing ^{95}Mo and ^{23}Na.

3.4.1. ^{11}B MAS NMR

Fig. 6 shows the experimental and simulated ^{11}B MAS NMR spectra for the 3Mo glass sample, pointing out the sites beneath trihedral and tetrahedral boron major contributions. The detailed identification of the four different contributions involved has been given elsewhere [30]. Before fitting, the central spectrum region corresponding to the central transition was corrected for the contribution from satellite transitions, mainly for tetrahedral boron because of its small quadrupolar interaction, as described in [30]. The correction was performed by removing the second spinning sideband from the central band, by subtracting from the spectrum the same spectrum shifted by the rotation frequency. All spectra were fitted with the values reported in Table 4. An example of the fit is reported in Fig. 6 for 3Mo glass.

The configuration in which tetrahedral boron atom is surrounded only by silicon atoms is generally found at about −2 or −3 ppm. The chemical shift then increases with the number of second-neighbor boron atoms by about 2–3 ppm per boron atom replacing a silicon atom [35]. The broadest peak corresponds to trihedral boron, attributable to the presence of two sites. First, boron surrounded by bridging bonds with boron atoms, generally referred to as boroxol rings ($BO_{3\ \mathrm{ring}}$) which could include other boron entities involving BO_3–O–BO_4 bonds, for example, and second, boron distributed in silicate units (generally referred as $BO_{3\ \mathrm{non\text{-}ring}}$) [35].

The proportions of tetrahedral boron (%BO_4) given by the fit of ^{11}B MAS NMR spectra are given in Table 3. Fig. 7 shows the ^{11}B MAS spectra for part of the glass series. There is a widening towards low chemical shifts. As previously demonstrated [30], it is suggested that this effect can be linked to the relative decrease of boron surrounded by three silicon atoms and one boron atom, compared to boron surrounded by four atoms of second neighbor silicon, with the increased MoO_3 content. In these conditions, a tetrahedral boron surrounded by other boron atoms could be more easily be transformed in a trihedral boron.

Table 3
Mass and volume percentages of crystalline phases, composition of surrounding glasses (% molar), percentage of BO_4 units determined by ^{11}B NMR and of NBOs in the glasses. Specific surface areas *SSA* of the glasses determined by krypton adsorption (after rinsing for $x = 2$ to 10), shape factor *Fs* and initial rate r_0 ($g \cdot m^{-2} \cdot d^{-1}$).

Glass	0Mo	0.5Mo	0.8Mo	2Mo	3Mo	4Mo	5Mo	10Mo
Crystalline phases (wt.%)	–	–	–	1.4	3.6	5.7	7.8	16.8
Crystalline phases (vol.%)	–	–	–	0.9	2.5	4.0	5.6	13.3
SiO_2	63.5	63.2	63.0	63.0	63.6	64.3	65.0	68.1
B_2O_3	16.9	16.8	16.8	16.8	16.9	17.1	17.3	18.1
Na_2O	19.6	19.5	19.4	18.8	18.1	17.2	16.3	12.4
MoO_3	–	0.5	0.8	1.4	1.4	1.4	1.4	1.4
BO_4 (%)	76.8	75.9	76.3	75.9	70.6	71.2	68.8	60.6
NBOs (%)	6.7	6.3	6.0	4.7	4.7	3.6	3.0	0.0
SSA ($m^2 \cdot g^{-1}$)	0.0480	0.0490	0.0595	0.0675	0.0750	0.0700	0.0790	0.0740
Fs	1.8	1.8	2.2	2.5	2.8	2.6	2.9	2.7
r_0 (Si) ($g \cdot m^{-2} \cdot d^{-1}$)	33.7	39.3	31.4	29.5	26.4	23.5	16.9	13.4

Fig. 8 shows the evolution of tetrahedral boron depending on the initial amount of molybdenum oxide added to the glass. It should be noted that the quantity of BO_4 remains stable up to 2% of MoO_3 then steadily decreases with the addition of more molybdenum oxide.

3.4.2. ^{29}Si MAS NMR

The increased silicate network polymerization with the addition of molybdenum oxide was clearly visible from the ^{29}Si MAS NMR spectrum shift towards the lower chemical shifts. The separation of Q^n species in the borosilicate glasses required care because, due to the wide distribution of environments which broadened the spectra, the silicon second neighbor boron atoms modified the chemical shift of the ^{29}Si peaks. Moreover, the Si second neighbor of BO_4 units in the form of Q^4 units caused a shift of approximately 10 ppm in the opposite direction to that of the polymerization increase [36]. However, BO_3 units do not cause chemical shift variations [37].

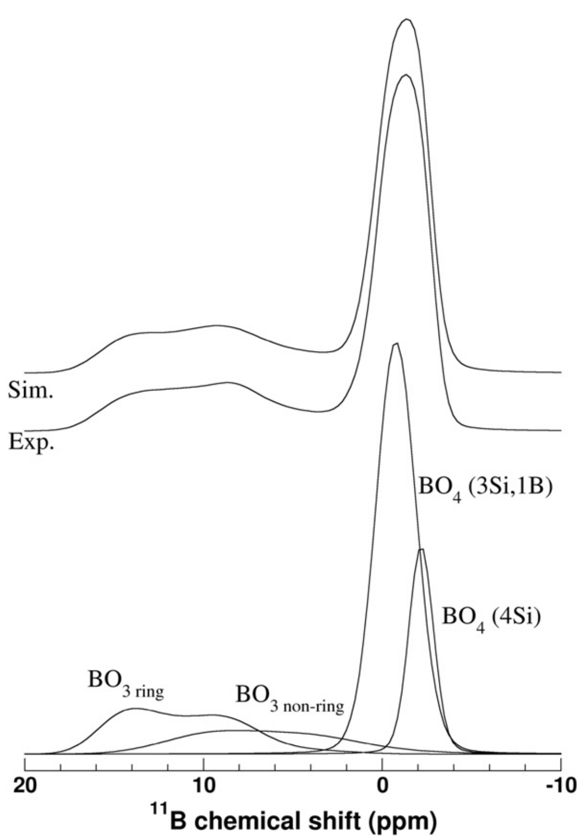

Fig. 6. Decomposition of ^{11}B MAS NMR spectrum for 3Mo glass.

In order to quantify the Q^n evolutions depending on the MoO_3 concentration, at least three contributions were necessary to fit the spectra (Fig. 9(a)): one contribution from the Q^3 species at −90 ppm and two contributions from the Q^4 species. The first at −98 ppm could be attributed to silicon atoms surrounded by three silicon atoms and one boron atom in tetrahedral coordination, and the second at −106 ppm in an environment entirely containing silicon and/or trihedral boron. The evolution of these different contributions is shown in Fig. 9(b). Because of the high degree of overlapping between the different NMR lines and the need to obtain the best fit, a variation of the Q^3 band had to be fixed, identical to that which would be expected given the number of NBOs shown in Table 3. It was considered that the NBOs only contribute to the Q^3 species. The spectra were fitted *simultaneously* with all the free parameters (chemical shift and peak width, intensity of the two Q^4 contributions). It should be noted that the use of two contributions only does not enable all the data to be correctly reproduced.

From the very good fitting agreement, it can be concluded that polymerization increases linearly with the addition of molybdenum, and that this occurs even for very low levels of MoO_3.

3.4.3. ^{95}Mo MAS NMR

Fig. 10 gives the ^{95}Mo MAS spectra for the 2Mo and 5Mo glasses, extracted from glass samples without separated macroscopic phases. The δ_{iso} of hydrated or non-hydrated sodium molybdate crystalline references are indicated in Fig. 10. The slight shift observed between the spectra of the two glasses is consistent with DRX-Rietveld results (Table 2) which showed that the addition of MoO_3 results in an increase of hydrated phases. The peak narrowing for the glass with the higher MoO_3 content could be explained a higher ordering around Mo, linked to the initiation of sodium molybdate crystallization.

It was possible to satisfactorily simulate the peaks using a Gaussian Isotropic Model [38], which enabled the NMR parameters listed in Table 5 to be obtained. In particular, they show the influence of increased crystallization on the decreasing quadrupolar interaction: the molybdenum sites probably become more symmetrical for the higher molybdenum oxide contents. Nevertheless, a weak variation in

Table 4
NMR parameters based on MAS spectrum analysis using NMR parameter distribution as described in [30]. Their mean values are shown together with standard deviation values, in brackets.

Site	δ_{iso} (ppm) − ($\sigma_{\delta_{iso}}$)	C_Q (MHz) − (σ_{C_Q})	η_Q − (σ_{η_Q})
$BO_{3\ ring}$	17.6 (1.3)	2.54 (0.1)	0.2 (0.1)
$BO_{3\ non\text{-}ring}$	13.7 (2.1)	2.54 (0.1)	0.2 (0.1)
BO_4 (1B, 3Si)	−0.3 (1.1)	0.55 (0.2)	0.6 (0.3)
BO_4 (0B, 4Si)	−1.8 (1.1)	0.21 (0.1)	0.6 (0.3)

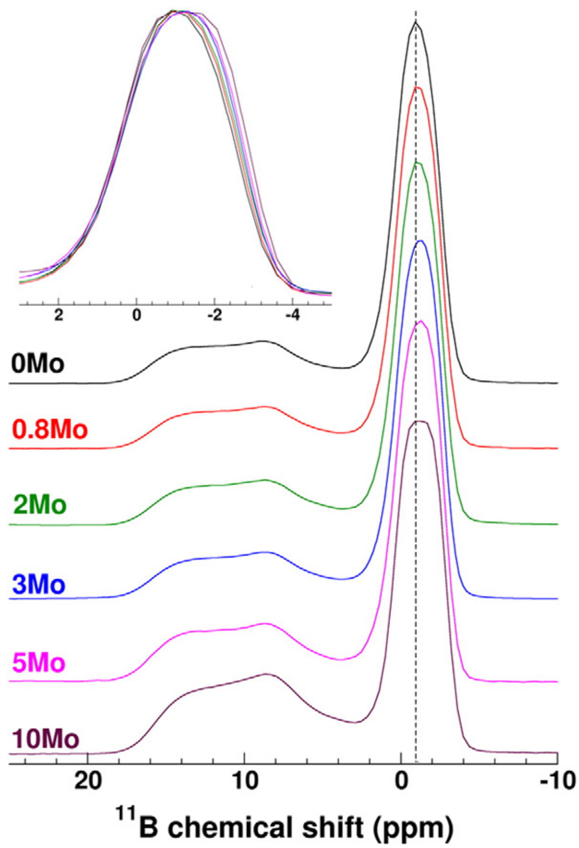

Fig. 7. ^{11}B MAS NMR spectra for *x*Mo glasses (spectra are normalized to the sample maximum height). Insets: enlarged view of BO_4 contribution.

chemical shift was observed, in agreement with the previous results [18].

3.4.4. DFT calculation of ^{23}Na NMR parameters

First-principle calculations of the ^{23}Na NMR parameters and geometry optimizations (atomic positions and cell parameters) for the crystalline compounds $Na_2Mo_2O_7$, α-Na_2MoO_4 and γ-Na_2MoO_4, Na_2SiO_3, α-$Na_2Si_2O_5$ and β-$Na_2Si_2O_5$ were performed using the GIPAW [39,42,43] method, as implemented in the VASP package [40] and the PBE GGA functional [41]. Before NMR parameter calculations, all structures were optimized using $3 \times 3 \times 3$ Monkhorst-Point k-grid and kinetic cutoff energy of 650 eV (isotropic chemical values converged by less than 0.1 ppm). As known for the PBE functionals,

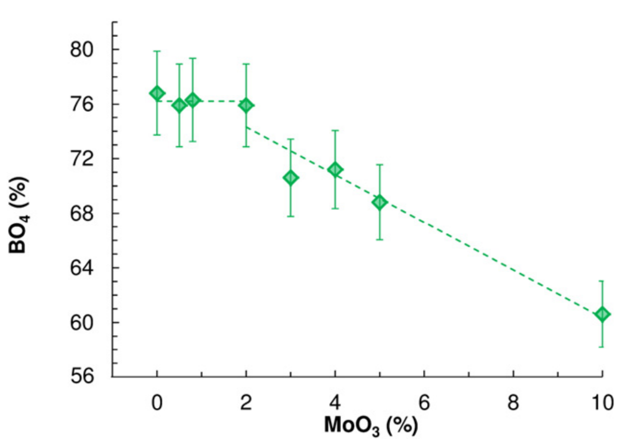

Fig. 8. Evolution of $\%BO_4$ depending on the glass molybdenum oxide content (errors bars represent 4% of uncertainties). Broken lines are a simple visual aid.

optimized unit cells were found to be overestimated by about 3.7% on average. As discussed in the Supplementary material, a better agreement between theoretical and experimental ^{23}Na isotropic chemical shifts values was observed for full optimization (FO) of the structure (atomic position and unit cell parameters) than when relaxing atomic positions only (APO), i.e., with experimental unit cell parameters. Therefore the FO data were used to predict to NMR data of the $Na_2Mo_2O_7$ and γ-Na_2MoO_4.

Procedures to relate the GIPAW outputs (magnetic shielding and electric field gradient tensors) to the experimentally measured quantities (isotropic chemical shift δ_{iso}, quadrupolar coupling constant C_Q and asymmetry parameter η) are described in references [45,46]. Briefly, the isotropic chemical shift δ_{iso} was obtained from the isotropic magnetic shielding σ_{iso} using the equation $\delta_{iso} = -\alpha(\sigma_{iso} - \sigma_{ref})$, where the parameters α and σ_{ref} were obtained from data shown in Fig. 11. These parameters were then used to predict the ^{23}Na parameters of the $Na_2Mo_2O_7$ phase for which, to the best of our knowledge, no NMR data has been reported to date. Predicted NMR parameters are given in Table 6.

3.4.5. ^{23}Na MQMAS NMR

Fig. 12(a) shows the ^{23}Na MQMAS NMR spectrum for the 5Mo glass containing separated phases dispersed in the vitreous matrix, as well as a phase separated macroscopically from the glass. It shows a narrow first contribution centered around 20 ppm on the isotropic dimension and a broader second contribution probably made up of several entities which could be fairly much distributed.

Fig. 12(b) enabled a clear identification of the species positioned on the isotropic dimension at 20 ppm as Na_2MoO_4, and probably a Na_2MoO_4, $2H_2O$ contribution to the broader line. The DFT calculation of the MQMAS spectrum for the $Na_2Mo_2O_7$ compound suggests that this phase should be ascribed to the weak peaks at −20 ppm in the isotropic dimension whereas the most intense peak at −8 ppm could be ascribed to a hydrated phase. Other hydrated sites could affect this extended contribution, which includes that of the residual glass.

Fig. 13 compares the projection of the two dimensions of the ^{23}Na MQMAS spectrum for 5Mo glass with those for the 0.8Mo glass. The narrow contributions linked to the crystallization of sodium molybdates appear distinctly on the 5Mo glass. Further, Fig. 13(b) shows a decrease in the average isotropic chemical shift with increased molybdenum oxide content. In agreement with previous studies, which correlate the increased isotropic chemical shift of ^{23}Na to the decreased Na–O bond distance [47,48], this observation shows that the Na–O distances are greater when a larger amount of sodium is found in the position of molybdenum charge compensator. This type of behavior is typically found when the Na is located near other network formers like tetrahedral boron or aluminum, acting as charge compensator [49].

3.5. Chemical durability

3.5.1. Initial alteration rate of the glasses

Alteration in the initial rate regime at pH 9 showed a congruent linear release of all the elements for the homogeneous glasses without crystallization (example of 0.8Mo glass, Fig. 14(a)).

For the glasses with separated phases containing sodium molybdate crystals ($x \geq 2$), the release was linear and incongruent (Fig. 14(b)). This behavior can be attributed to a sodium and molybdenum release initiated by the dissolution of sodium molybdates present in the glass. Because of their very high solubility [50,51], sodium molybdates macroscopically separated from the glass or dispersed within the separated phases on the glass grain surface are preferentially dissolved as soon as they are in contact with the solution, as illustrated by Fig. 1.

To avoid the sodium molybdate dissolution, the glass powder was rinsed to remove macroscopic or microscopic surface phases. The specific surface area of washed and unwashed glass powders was compared. Due to the dissolution of molybdenum phases having a

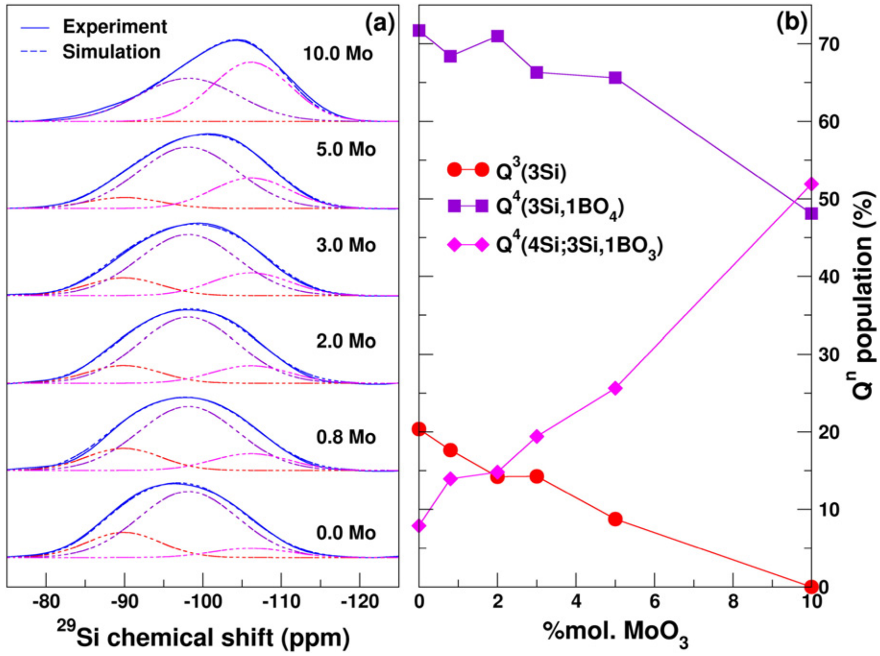

Fig. 9. (a) ^{29}Si MAS NMR spectra and (b) evolution of the Q^3 and Q^4 species depending on the MoO_3 content (two different species of Q^4 were considered, see text).

high roughness (Fig. 2(d)), a sharp decrease was observed for glasses with a macroscopic separated phase ($x \geq 3$). After rinsing, the shape factor ($F_s = SSA$ / geometric surface, see Table 3) of powders was slightly higher than those obtained for homogeneous glasses ($x \leq 0.8$). These variations can be attributed to the cavities scattered over the surface of the surrounding glass. Washed glasses were then leached under initial alteration rate conditions. The dissolution was therefore congruent (Fig. 14(c)) and the pH did not change during the test. Obtained on at least three experiments for each glass, the average initial alteration rates are listed in Table 3.

3.5.2. Alteration in rate drop regime

To take into account the long term influence which pH changes caused by sodium molybdate dissolution could have (MoO_3 releases lead to a solution acidification according to the following reaction: $MoO_3 + 2OH^- \rightarrow MoO_4^{2-} + 2H_2O$), unwashed glasses were used for these experiments. To maintain a SSA/V ratio of 1500 m^{-1} for the glasses with $x \geq 2$, the mass m of crystallized glass to be added was determined based on the molybdenum phase content and using the specific surface areas of the surrounding glasses (Table 3).

For example in Fig. 15(a), representing 3Mo glass alteration, equal normalized mass losses of boron, molybdenum and sodium are obtained, whereas the silicon release is much lower. For all the glasses, silicon retention of about 90% was observed. SEM images of the glasses after 35 days of alteration confirmed that the alteration layer was exclusively composed of silicon (Fig. 15(c) and (d)). The thickness of the alteration layer observed by SEM ($\approx$3 μm) is in good agreement with boron release in solution (Fig. 15(a)) giving after 28 days of alteration an equivalent thickness of 3.1 μm.

The evolutions of normalized mass losses for boron and of the pH versus time for the different glasses are shown in Fig. 16. Whether homogeneous or crystallized, all the glass series had alteration kinetics of similar magnitude after varying periods of rate drop. The quantity of altered glass in the vitreous part decreased considerably as the molybdenum oxide content increased. The pH became lower with the higher MoO_3 content, with a difference of one unit between the glasses at the lower and upper limits of the series studied ($x = 0$ and $x = 10$).

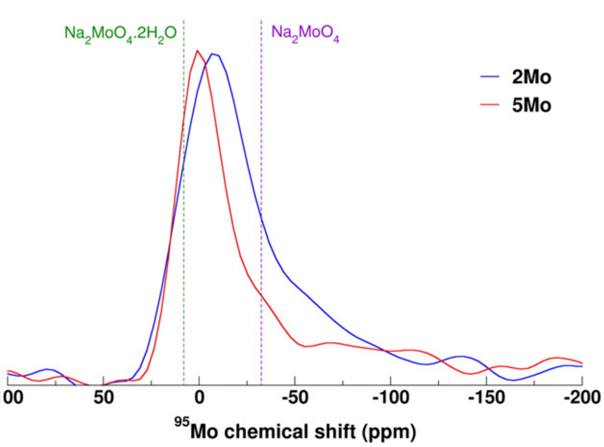

Fig. 10. ^{95}Mo MAS NMR spectra for the 2Mo and 5Mo glasses. The dashed lines indicate the resonance δ_{iso} for hydrated sodium molybdate and non-hydrated sodium molybdate.

4. Discussion

4.1. Influence of phase separation on the sodium borosilicate glass microstructure and structure

The processes of phase separation which take place in sodium borosilicate glasses may have a major influence on the microstructure [52–54] and the structure of the vitreous network [22,35]. The microstructural characterization highlights the strong influence of the MoO_3 concentration initially added to the glass on the nature and the quantity

Table 5
NMR parameters for the ^{95}Mo sites in the 2Mo and 5Mo glasses.

	2Mo	5Mo
δ_{iso} (ppm)	13.5	15.7
C_Q (MHz)	2.3	2.0
η	0.6	0.6

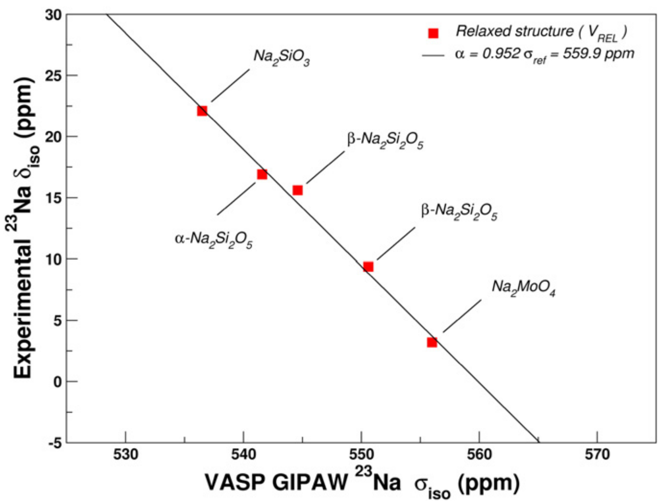

Fig. 11. Experimental ^{23}Na isotropic chemical shift δ_{iso} versus the theoretical isotropic shielding σ_{iso} (obtained using DFT-GIPAW VASP calculations). The linear regression $\delta_{iso} = -\alpha(\sigma_{iso} - \sigma_{ref})$ is shown.

Table 6
Predicted ^{23}Na parameters with GIPAW calculations of $Na_2Mo_2O_7$ and γ-Na_2MoO_4 using relaxed structures.

Phase	σ_{iso}(ppm)	C_Q(MHz)	η	δ_{iso}(ppm)
$Na_2Mo_2O_7$ (Na1)	575.7	2.4	0.3	−15.0
$Na_2Mo_2O_7$ (Na2)	570.8	1.5	0.6	−10.3
γ-Na_2MoO_4	574.1	4.5	0.5	−13.5

of phases formed. Homogeneous when the MoO_3 concentration is lower than 1.4% molar, the glasses contain micrometric spherical phases (Fig. 3) when MoO_3 reaches 2% molar, and macroscopic phases for higher concentrations. After cooling, these phases mainly contain Na_2MoO_4 and $Na_2Mo_2O_7$-type crystalline phases. The surrounding vitreous matrix is thus progressively depleted in molybdenum and sodium, which become concentrated in the crystalline separated phases. This type of behavior can be compared with the studies referenced by Gutzow [55], which highlight the different processes of possible crystallization in demixed glasses. From the microstructures observed, it is thus possible to suggest that phase separation is produced by a first stage of nucleation and growth, followed by the crystallization of separated phases enriched solely in MoO_3 and Na_2O.

The composition variations caused by phase separation lead to a modification of the borosilicate network structure of the surrounding glasses through a charge compensation mechanism of the molybdate entities in two stages: a compensation solely by the sodium ions coming from the Si-NBO entities for low MoO_3 levels (up to 2% molar), then coming from BO_4^- entities above this content. Polymerization increase of the silicate network takes place as soon as even a very low quantity of MoO_3 is added to the glass. This effect, suggested in the case of silicate [48] and borosilicate glasses [23] was checked and quantified here over a wide range of variations of MoO_3 content.

4.2. Influence of glass structure on the initial alteration rate regime

The initial alteration rate regime is the most likely to be directly influenced by the structural organization of a glass, in particular through polymerization of the vitreous network [56], the insertion of network formers of different natures [57], or the distribution of network modifiers [58]. These structure-reactivity correlations are however often complex to demonstrate, as numerous factors vary simultaneously when the chemical composition of the glass changes [32,59]. Moreover, the same element can have opposite effects depending on its concentration in the glass. For example, the addition of multivalent modifier cations like alkaline-earths can, at low levels, limit the hydrolysis of the network by mobilizing NBOs [58,60] or, on the contrary, at high levels it can increase this by a predominantly depolymerizing effect [60].

The limited number of parameters studied here, i.e., (i) addition of MoO_3 to substitute for all the elements in order to maintain the Si/B, Si/Na, B/Na ratios constant, (ii) alteration at a constant pH, enables more direct relationships between the structural parameters and the alteration kinetics to be obtained. Fig. 17 shows a clear correlation between the decrease in the initial alteration rate of the surrounding glass and the drop in the number of NBOs.

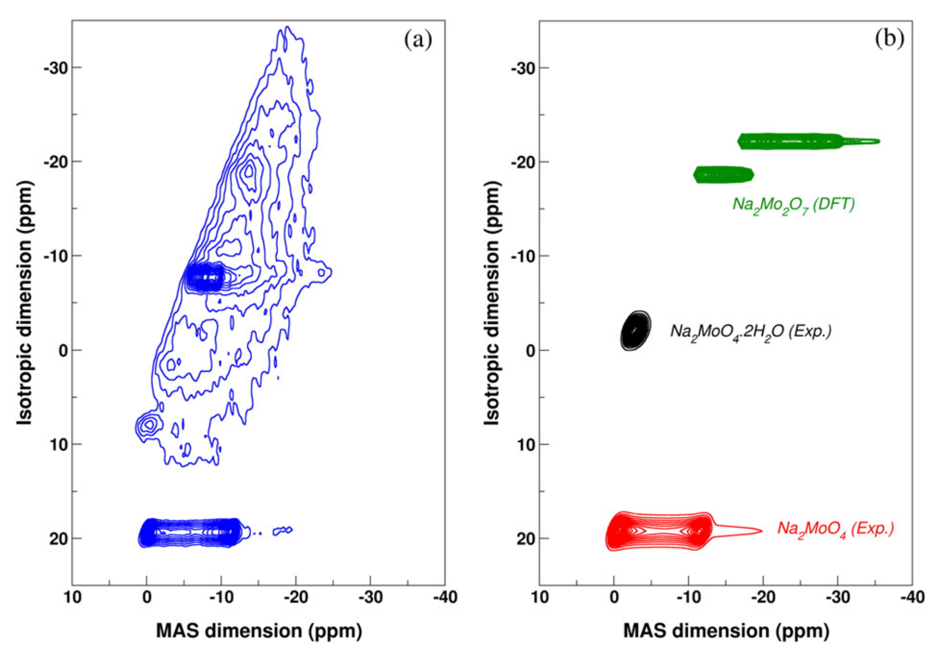

Fig. 12. (a) ^{23}Na MQMAS NMR spectrum for the 5Mo glass, (b) ^{23}Na MQMAS NMR spectra for Na_2MoO_4, $2H_2O$ and Na_2MoO_4 compounds and spectrum calculated by DFT for $Na_2Mo_2O_7$.

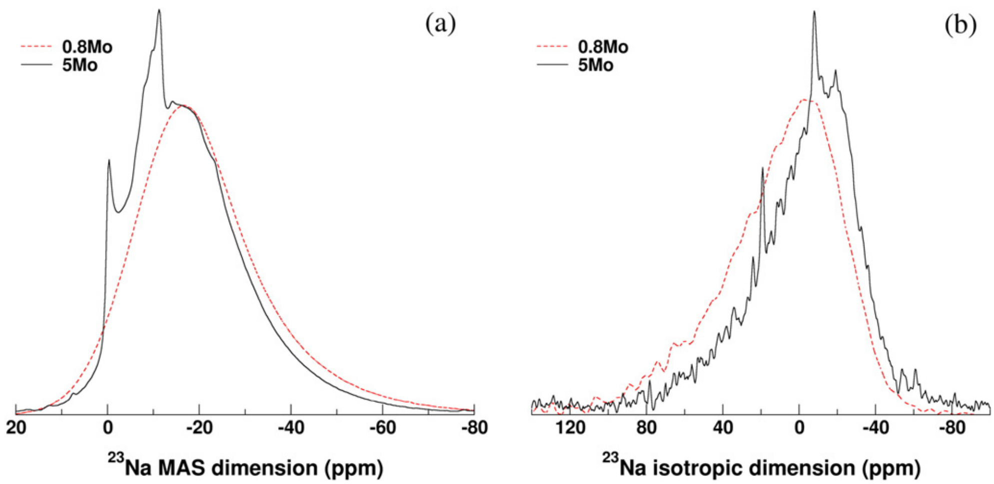

Fig. 13. Projections on the MAS dimension (a) and isotropic dimension (b) of the ^{23}Na MQMAS NMR spectra for 0.8 Mo and 5Mo glasses.

In agreement with thermodynamic data [61], it can be suggested that this behavior could be extrapolated to higher MoO_3 contents. The miscibility gap at 1473 K represented in the ternary SiO_2–Na_2O–MoO_3 shows the continuity of the phase separation phenomenon between the silicate network and the molybdates. In particular, it indicates that this phenomenon can occur for MoO_3 concentrations less than or equal to 20% molar. The initial dissolution rate of the vitreous part of the material thus should continue to decrease. These results point out the sensitivity of the initial alteration rate measurement to the silicate network polymerization.

4.3. Influence of the nature of the separated phases on the rate drop regime

For higher reaction progress, the glasses were altered until the normalized mass losses reached a plateau, indicating the end of the rate drop regime (Fig. 16(a)). The chemical composition of the glass, modified by increasing amounts of molybdenum, and pH imposed by the amount of dissolved molybdenum phase in the early stages influence the time to reach the steady state. To favor comparisons between the different glasses, the alteration behavior can be described by a first order kinetic law [62,63] using Eq. (4):

$$NL_i(t) = NL_i^*\left(1-e^{\frac{-t}{\tau}}\right) \tag{4}$$

where $NL_i(t)$ is the normalized mass loss for element i at time t, NL_i^* is the normalized mass loss for element i reached at the end of the alteration period, and τ is the time constant for element i at the end of which the steady state is reached.

The parameters of Eq. (4) obtained for the boron normalized mass loss for each of the xMo glasses are listed in Table 7. The comparison of the calculated normalized mass loss values, NL^*_B *fit*, with the experimental values, NL^*_B exp, shows a good agreement.

The calculation of the $\tau_{B\ (0Mo)}/\tau_{B\ (xMo)}$ ratio (Table 7) enables quantification of the impact of the pH on the time necessary to reach the steady state.

Thus, Fig. 18 highlights three regimes depending on the pH: (i) for glasses with a MoO_3 content lower than 2%, pH variations are weak and have no impact on the time to reach the steady state; (ii) above 2%, the solution pH is strongly impacted – both by the macroscopic sodium molybdate phase dissolution and the low surrounding glass dissolution which results in a low sodium release and thus limits the pH increase – and a rapid evolution can be seen, with the steady state reached up to five times faster than for the 5Mo glass; (iii) in spite of the pH variations caused by the higher phase dissolution rate for higher

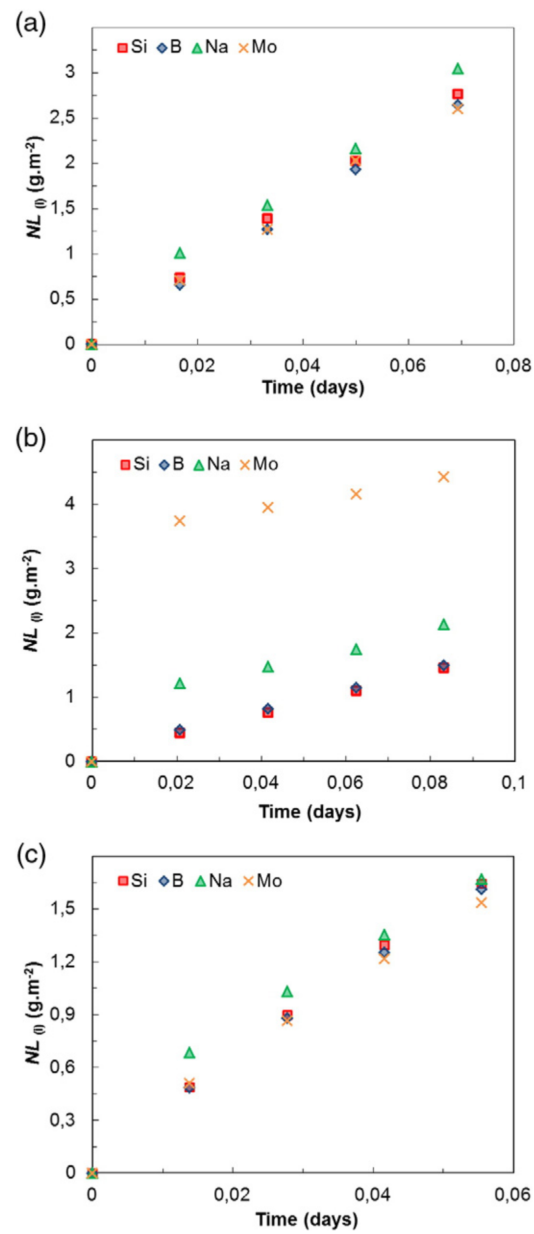

Fig. 14. Standardized mass losses for the elements Si, B, Na and Mo over time in the glasses (a) 0.8Mo, (b) 4Mo and (c) rinsed 4Mo.

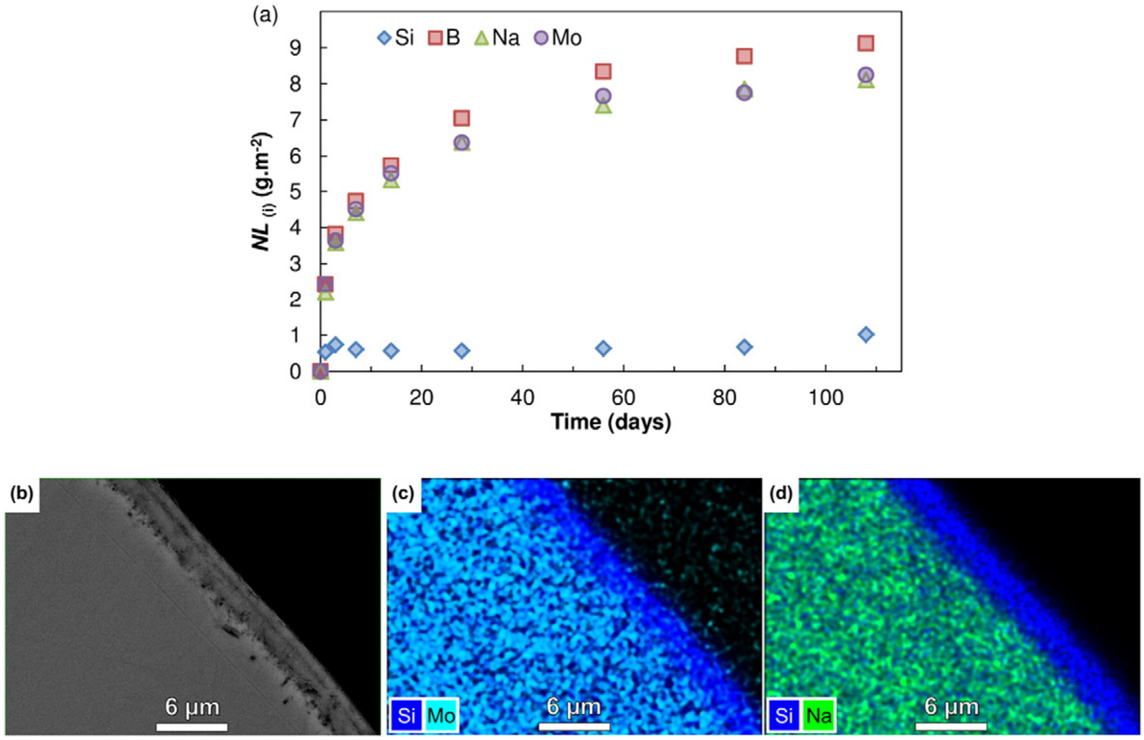

Fig. 15. (a) Normalized mass losses of boron, molybdenum, silicon and sodium from the 3Mo glass after removing initial release of Na and Mo due to sodium molybdates dissolution and (b) SEM images of the alteration layer profile for 3Mo glass after 35 days of alteration, (c) and (d) corresponding EDX mappings.

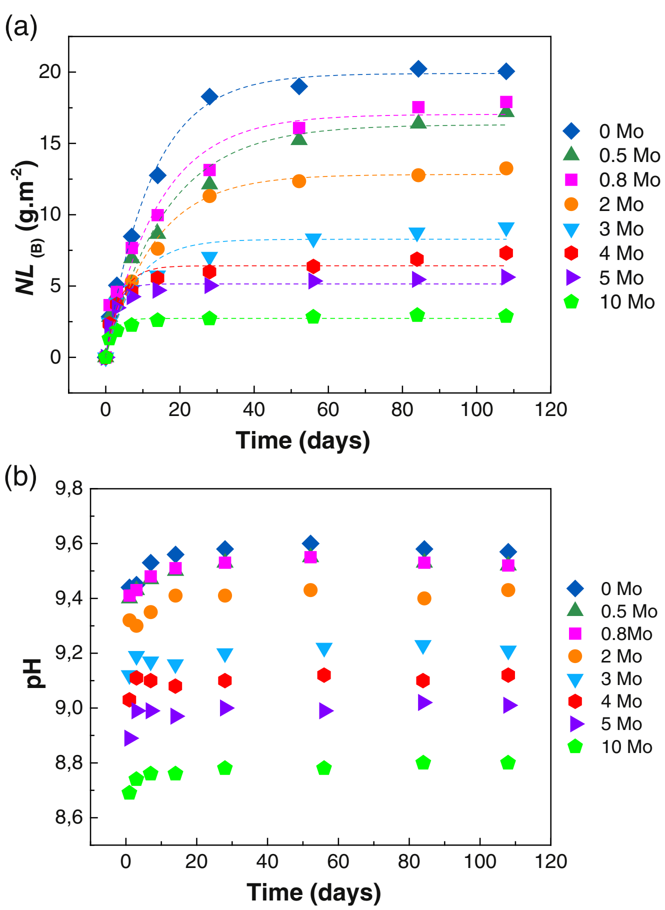

Fig. 16. Evolution with time for (a) normalized mass losses for boron and (b) the pH of *x*Mo series glasses.

MoO_3 contents, the time constant seems to be close to a maximum for the 10Mo glass.

The silicon concentration in the solution at the steady state can be compared to that imposed by the dissolution of the amorphous silica. Such a comparison is justified as the alteration layer controlling the glass alteration is mainly composed of silicon (Fig. 15). Based on amorphous silica dissolution calculations with the geochemical speciation code CHESS [64], its solubility limit depending on the pH was determined. The comparison of the silicon concentrations measured in the solutions at the end of the leaching experiment with the solubility limit for amorphous silica (Fig. 18(b)) shows that the *x*Mo glass alteration in the rate drop regime is highly dependent on the pH. The decrease of pH with the addition of MoO_3, and therefore with sodium molybdate dissolution as well (Fig. 16(a)), leads to saturation being reached more rapidly. It controls the glass alteration by

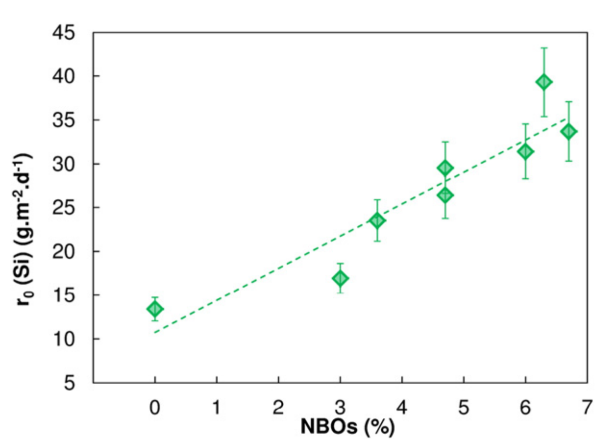

Fig. 17. Evolution of the initial dissolution rate depending on the percentage of NBOs (errors bars represent 10% of uncertainties).

Table 7
Alteration parameters of the first order kinetics law (τ_B and NL^*_B *fit* according to Eq. (4)), boron normalized mass loss at the end of the experiment (NL^*_B exp), ratio $\tau_{B\ (0Mo)}/\tau_{B\ (xMo)}$ for the *x*Mo glasses, and silicon concentrations in solution at 108 days of alteration ($[Si]_{108\ d}$).

Glass	0Mo	0.5Mo	0.8Mo	2Mo	3Mo	4Mo	5Mo	10Mo
τ_B (days)	12.4 ± 0.9	16.3 ± 2.7	14.3 ± 2.5	13.2 ± 1.6	8.0 ± 2.9	3.9 ± 0.9	2.5 ± 0.5	2.2 ± 0.4
NL^*_B *fit* ($g \cdot m^{-2}$)	19.9 ± 0.4	16.3 ± 0.8	17.0 ± 0.8	12.8 ± 0.4	8.3 ± 0.5	6.4 ± 0.3	5.1 ± 0.2	2.7 ± 0.1
NL^*_B exp ($g \cdot m^{-2}$)	20.1	17.2	17.9	13.25	9.1	7.3	5.6	2.9
$\tau_{B\ (0Mo)}/\tau_{B\ (xMo)}$	–	0.8	0.9	0.9	1.6	3.2	4.9	5.5
$[Si]_{108\ d}$ ($mg \cdot L^{-1}$)	871	787	1006	442	377	338	287	248

triggering the rate drop earlier, which means smaller quantities of altered glass for high MoO_3 content.

The $\tau_{B\ (0Mo)}/\tau_{B\ (xMo)}$ ratio stabilization below pH 9 (Fig. 18(a)) can be explained by the low variation in the amorphous silica solubility for this range of pH (Fig. 18(b)).

4.4. Influence of the phase separation mode on glass alteration

The behavior of demixed glasses during aqueous alteration is strongly correlated with the nature of the separated phases. Because of the variability of the compositions and the morphologies possible, this dependence is generally difficult to predict. Generic behaviors can however be identified depending on the phase separation mode, i.e., whether the glasses are obtained by spinodal decomposition (two interconnected phases in equivalent volume proportions) or by nucleation and growth (weakly connected separated phases, dispersed within the vitreous matrix). According to the classification proposed by Tomozawa [65] (i.e., type A glass: connected phases with variable durability, type B glass: separated phases dispersed within a surrounding vitreous phase, type C glass: separated phases dispersed within a mainly durable surrounding vitreous phase), the behavior of glasses demixed by the addition of MoO_3 can be considered similar to that of type C. By triggering the mobilization of two sodium atoms per atom of molybdenum, the addition of MoO_3 tends to polymerize the silicate network and thus to increase the surrounding vitreous phase durability.

Because of the low phase connectivity (due to the process of nucleation and growth) and their low proportion compared to the surrounding glass (which represents between 83.2 and 98.6% of the mass), the reactive surface variations caused by the dissolution of sodium molybdate phases (leading to cavity formation) are small and have little impact on alteration. The opposite situation would be observed for a system demixed by spinodal decomposition, as once the slightly soluble phase has been dissolved it leaves a very large reactive surface, leading to a very fast alteration of the material.

5. Conclusions

Until now, the chemical durability of vitreous matrices containing molybdenum oxide had been approached from complex borosilicate matrices [10,66]. The approach described above, with characterization at different scales on a model glass, has enabled a mechanistic understanding of the chemical durability of heterogeneous matrices.

This work has particularly focused on the influence of liquid–liquid phase separation caused by the addition of MoO_3 to sodium borosilicate glasses, for the glass microstructure, structure and chemical durability. Structural analyses showed the occurrence of a nucleation and growth process in the separated phases enriched in sodium and molybdenum oxides dispersed within a surrounding vitreous matrix. By mobilizing two atoms of sodium, the addition of molybdenum oxide leads first to a decrease in the number of non bridging oxygens, and then in the number of tetrahedral borons as the quantity of separated phases increases.

These structural modifications mean that the surrounding glass has a higher chemical durability in the different alteration regimes studied. In the initial rate regime, the leaching behavior can be directly correlated with the quantity of NBOs. The dissolution of highly hydrosoluble sodium molybdate-type crystallized separated phases has no impact on the overall glass behavior. In the rate drop regime, the dissolution of sodium molybdate highlights the contribution of the nature and the composition of the separated phases to the alteration solution chemistry.

The methodology developed in this study highlights two major issues for the chemical durability optimization of partially crystalline conditioning matrices containing high levels of molybdenum oxide:

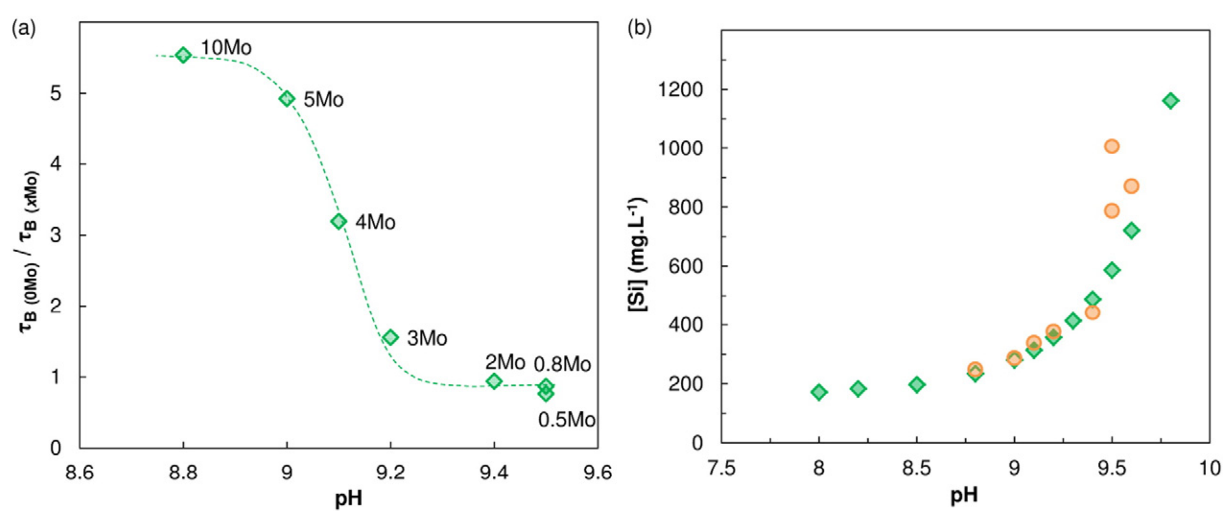

Fig. 18. (a) Comparison of the times necessary to reach the steady state depending on the pH represented by the ratio $\tau_{B\ (0Mo)}/\tau_{B\ (xMo)}$. This shows that the rate drop is reached 5 times more quickly for the 5Mo glass than for the 0.5Mo sample. Broken line is a simple visual aid. (b) Evolution depending on the pH for: ◇ the silicon concentration corresponding to the solubility limit for the amorphous silica and ○ the silicon concentration, at 108 days of alteration.

the surrounding matrix composition and the phase separation mode. In parallel, the nature of the crystalline phases arising from phase separation by nucleation and growth must be taken into account.

While it is possible today to orient the crystallization of durable phases [50,67] such as powellite ($CaMoO_4$) [18,22,24], this study has shown that for high MoO_3-content glasses, this should not be the only criteria taken into account. The durability of such heterogeneous materials also depends on the chemical composition and structure of the surrounding matrix as well as the environment chemistry, which can be partly imposed by the dissolution of demixed phases.

Acknowledgments

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Appendix A. Supplementary data

Supplementary data to this article can be found online at http://dx.doi.org/10.1016/j.jnoncrysol.2015.07.001.

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